Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

Hannah Kettle MATHS TUTORIALS

FOR FULL VIDEO SOLUTIONS SCAN THE QR CODE

youtube.com/@hannahkettlemaths



Half Past Paper – Week 7

Total Marks

Q	Topic		Sparx Code
1a	Probabilities	2	U510,U683
1b	Probabilities	2	U510,U683
3	Finding the highest common factor (HCF)	2	U529
5	Combining transformations	3	U766
7	Area and perimeter of rectangles and compound shapes problems	4	U993, U934
9	Multiplying with mixed numbers	3	U224
11a	Calculating the median	2	U456
11b	Choosing suitable averages and solving problems	1	U717
11c	Choosing suitable averages and solving problems	1	U717
13	Writing algebraic proofs	2	U582
15	Finding the volume of composite shapes	4	U543
17	Calculating with ratios and algebra, Constructing and solving quadratic equations	4	U676
19i	Completing the square	2	U397
19ii	Finding the turning point of a quadratic graph by completing the square	1	U769
21a	Finding inverse functions	2	U996
21b	Finding composite functions	5	U448
	Total	40	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 There are only blue cubes, red cubes and yellow cubes in a box.

The table shows the probability of taking at random a blue cube from the box.

Colour	blue	red	yellow
Probability	0.2		

The number of red cubes in the box is the same as the number of yellow cubes in the box.

(*a*) Complete the table.

(2)

There are 12 blue cubes in the box.

(*b*) Work out the total number of cubes in the box.

.....

(2)

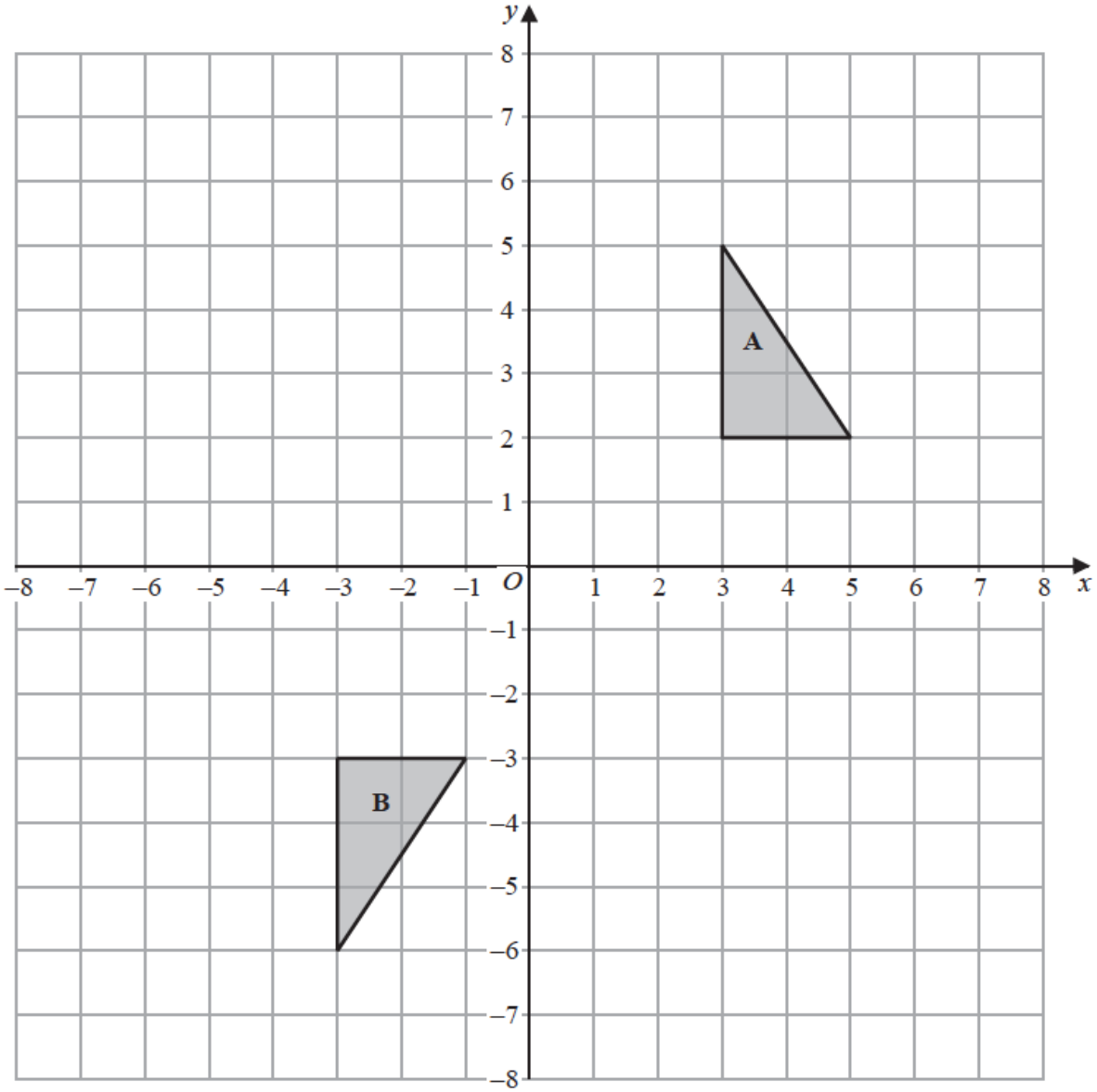
(Total for Question 1 is 4 marks)

3 Find the highest common factor (HCF) of 72 and 90

.....

(Total for Question 3 is 2 marks)

5



Shape **A** can be transformed to shape **B** by a reflection in the x-axis followed by a translation $\begin{pmatrix} c \\ d \end{pmatrix}$

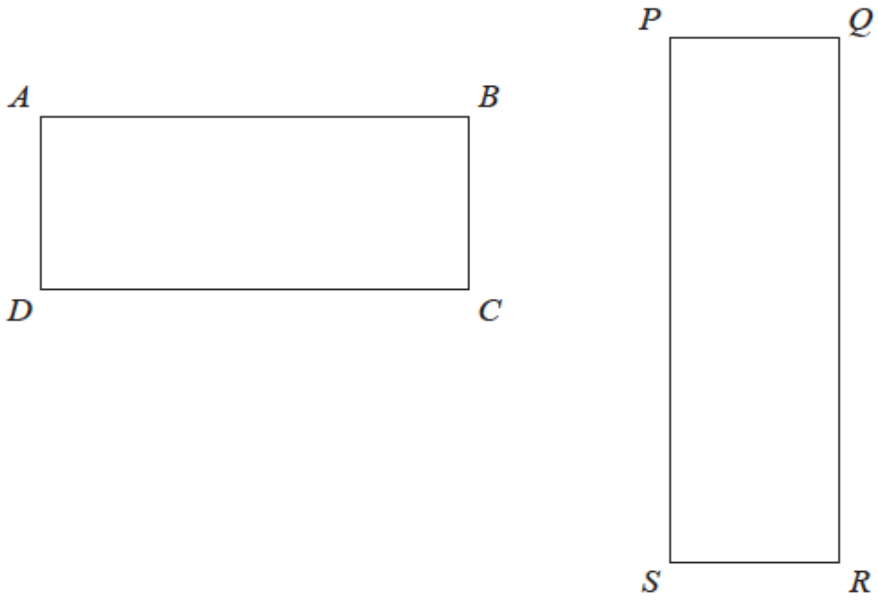
Find the value of c and the value of d.

c = ...

d = ...

(Total for Question 5 is 3 marks)

7 Here are two rectangles.



$QR = 10$ cm
$BC = PQ$

The perimeter of $ABCD$ is 26 cm
The area of $PQRS$ is 45 cm^2

Find the length of AB.

..... cm

(Total for Question 7 is 4 marks)

9 Work out $3\frac{1}{2} \times 1\frac{3}{5}$

Give your answer as a mixed number in its simplest form.

.....

(Total for Question 9 is 3 marks)

11 A bus company recorded the ages, in years, of the people on coach A and the people on coach B.

Here are the ages of the 23 people on coach A.

41	42	44	48	52	53	53	53	56	57	57	59
60	61	63	64	64	66	67	69	74	77	79	

(*a*) Complete the table below to show information about the ages of the people on coach A.

Median	
Lower quartile	
Upper quartile	
Least age	41
Greatest age	79

(2)

Here is some information about the ages of the people on coach B.

Median	70
Lower quartile	54
Upper quartile	73
Least age	42
Greatest age	85

Richard says that the people on coach A are younger than the people on coach B.

(*b*) Is Richard correct?
You must give a reason for your answer.

.....

.....

.....

(1)

Richard says that the people on coach A vary more in age than the people on coach B.

(*c*) Is Richard correct?
You must give a reason for your answer.

.....

.....

.....

(1)

(Total for Question 11 is 4 marks)

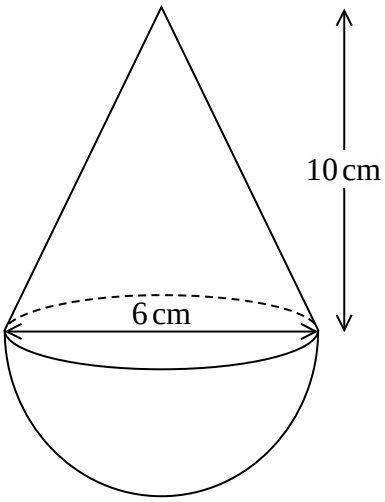
---

13 Given that n can be any integer such that $n > 1$, prove that $n^2 - n$ is never an odd number.

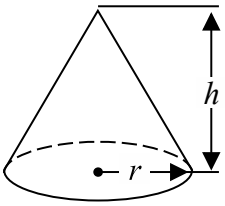
(Total for Question 13 is 2 marks)

---

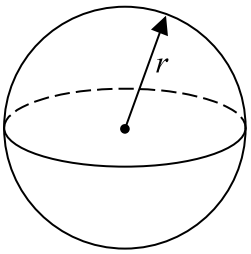
15 The diagram shows a solid shape.
The shape is a cone on top of a hemisphere.



Volume of a cone = $\frac{1}{3}\pi r^2 h$



Volume of a sphere = $\frac{4}{3}\pi r^3$



The height of the cone is 10 cm.
The base of the cone has a diameter of 6 cm.
The hemisphere has a diameter of 6 cm.

The total volume of the shape is $k\,\pi$ cm^3, where k is an integer.

Work out the value of k.

k = ...

(Total for Question 15 is 4 marks)

17 Given that

$$x^2 : (3x + 5) = 1 : 2$$

find the possible values of x.

.....

(Total for Question 17 is 4 marks)

19 Given that $x^2 - 6x + 1 = (x - a)^2 - b$ for all values of x,

(i) find the value of a and the value of b.

$a =$ .....

$b =$ .....

(2)

(ii) Hence write down the coordinates of the turning point on the graph of $y = x^2 - 6x + 1$

(..... , ...)

(1)

(Total for Question 19 is 3 marks)

---

21 The functions f and g are such that

$$f(x) = 3x - 1 \qquad \text{and} \qquad g(x) = x^2 + 4$$

(*a*) Find $f^{-1}(x)$

$f^{-1}(x) =$ .....

(2)

Given that $fg(x) = 2gf(x)$,

(*b*) show that $15x^2 - 12x - 1 = 0$

(5)

(Total for Question 21 is 7 marks)

TOTAL FOR PAPER IS 40 MARKS